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Bank Term Deposit Subscription Prediction

Machine Learning Approach to Direct Marketing

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<https://github.com/Judyluo1217/Bank-Term-Deposit-Subscription-Prediction.git>



Intro

01 Task

Predict client subscription (Yes / No)

02 Importance

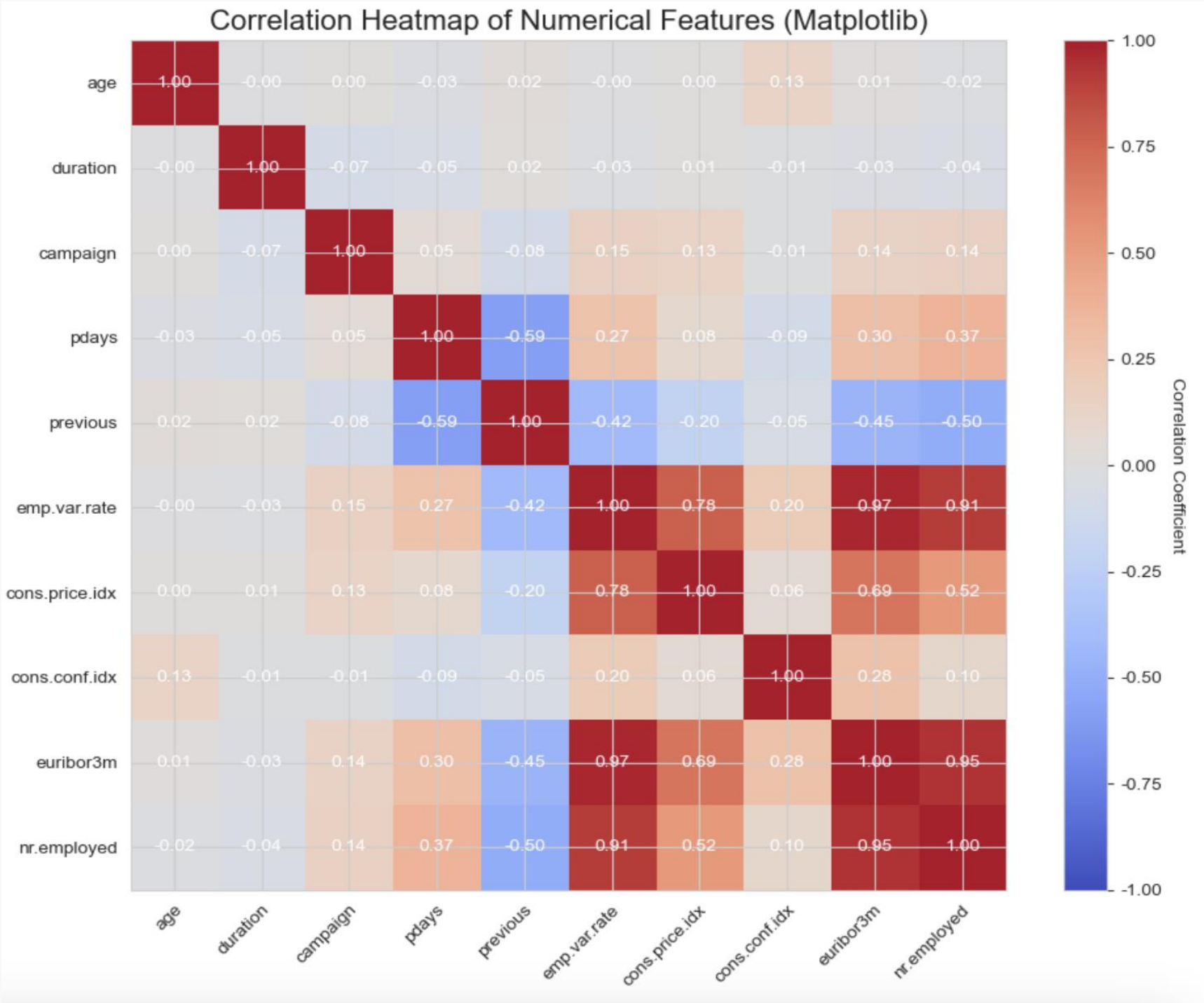
Optimize marketing campaigns, improve customer targeting

03 Data

- Bank Marketing Dataset (UCI ML Repository)
- 41,188 samples with 20 features

04 EDA & Preprocessing

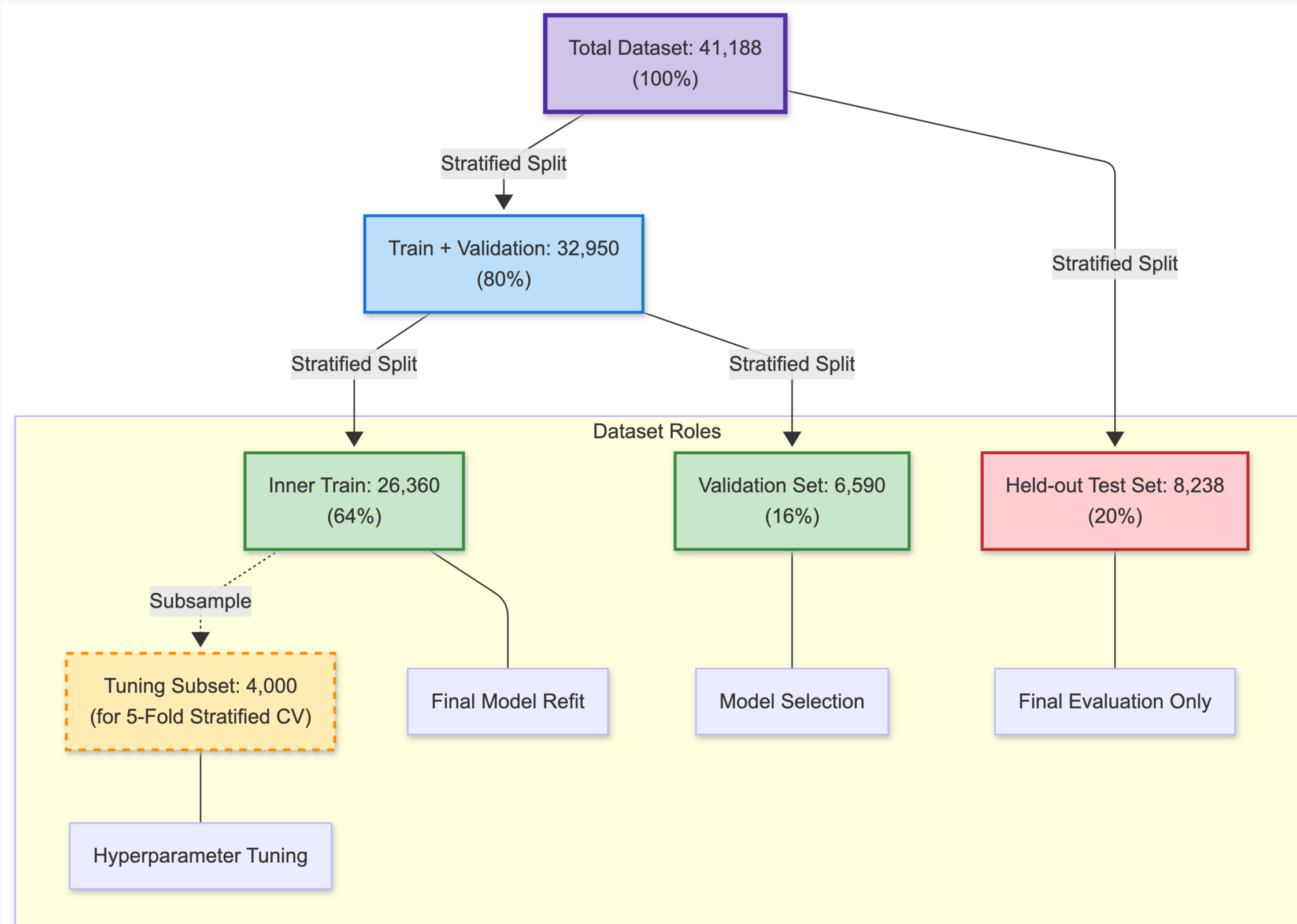
- Highly imbalanced: ~11% positive class (subscribed)
- Drop 'duration' ---- data leakage
- Drop emp.var.rate and euribor3m ---- high correlation (>0.9) with nr.employed
- Only have categorical missing values ('unknown' → NaN) before preprocessing [Ordinal encoding & One-hot encoding & Standardization]



Feature	R2 (Explained by others)	Correlation with y
emp.var.rate	0.743968	-0.29369
euribor3m	0.662549	-0.303174
nr.employed	0.506264	-0.349729



Data Split & CV Pipeline



Why this approach?

- Prevents data leakage (test set untouched until final evaluation)
- Reduces computational cost (subsampling for hyperparameter tuning)
- Maintains class balance throughout

CV pipeline

- GridSearchCV: 5-fold
- Metric: F1-Macro
- Model Parameter:
class_weight='balanced' &
scale_pos_weight for XGBoost

Models & Hyperparameters



Selected Champion: Random Forest

- Best validation F1-Macro: 0.7101 & Test F1-Macro: 0.7182 (no overfitting or underfitting)
- Best parameters: {max_depth: 15, n_estimators: 100, max_features: 'log2', min_samples_split: 20}

Model Key	Hyperparameters Tuned	Type	Train F1	Validation F1
Logistic Regression	C: np.logspace(-3, 3, 7) penalty: (L1/L2)	Linear	0.6222 ± 0.020	0.6316
SVM	C: [0.1, 1, 10, 100] gamma: [0.0001, 0.0005, 0.001] kernel: ['rbf']	Nonlinear	0.6566 ± 0.034	0.5857
KNN	n_neighbors: [2, 3, 5, 7] weights: ['uniform', 'distance'] p: [1, 2]	Nonlinear	0.6421 ± 0.045	0.6487
Random Forest	max_depth: [10, 15, 30, 50] n_estimators: [100, 200, 300] max_features: ['sqrt', 'log2', 0.3, 0.5, 0.7] min_sample_split: [10, 20, 40]	Nonlinear	0.6815 ± 0.030	0.7101
XGBoost	early_stopping_rounds=50 learning_rate: [0.001, 0.005, 0.01] max_depth: [5, 7, 9, 11] subsample: [0.5, 0.6, 0.7, 0.8] reg_lambda: [0, 0.01, 0.05, 0.1]	Nonlinear	0.6784 ± 0.020	0.7022

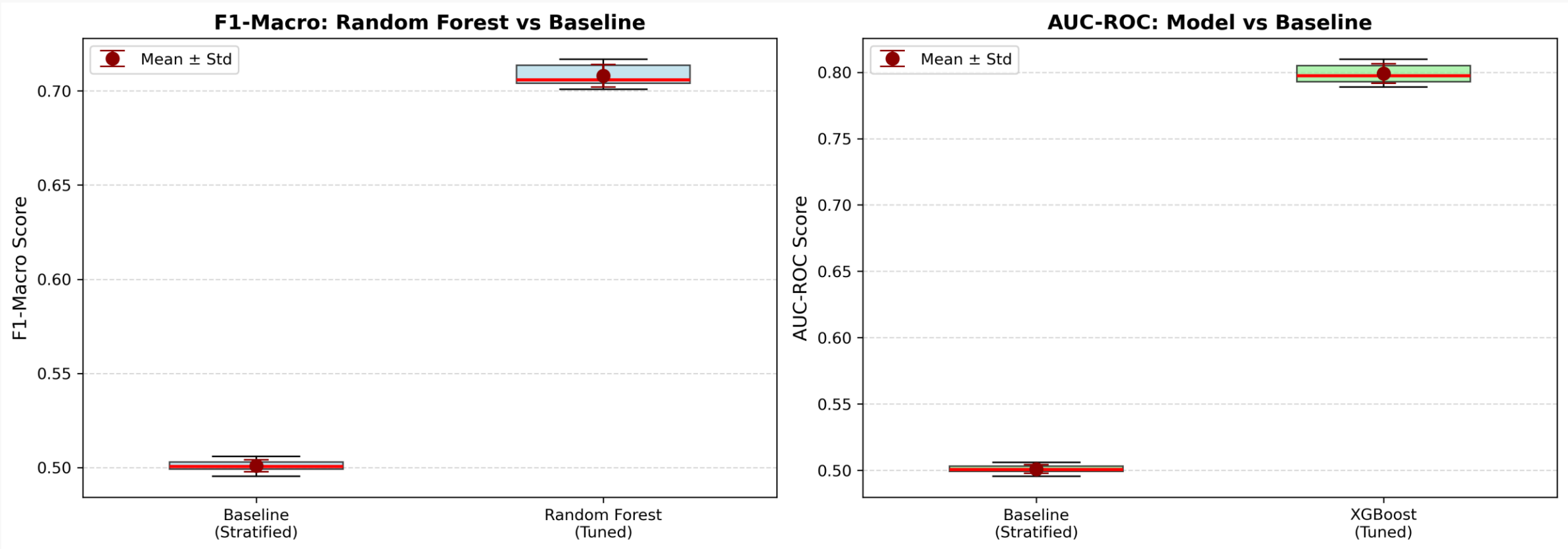
Uncertainty Analysis



Robustness Analysis (10 different random seeds)

Metric	Mean	Std	Min	Max
RF F1-Macro	0.708	0.006	0.7008	0.7168
RF AUC-ROC	0.7989	0.0073	0.789	0.8098
Baseline F1	0.501	0.0033	0.4954	0.5059
Baseline AUC	0.501	0.0033	0.4954	0.5058

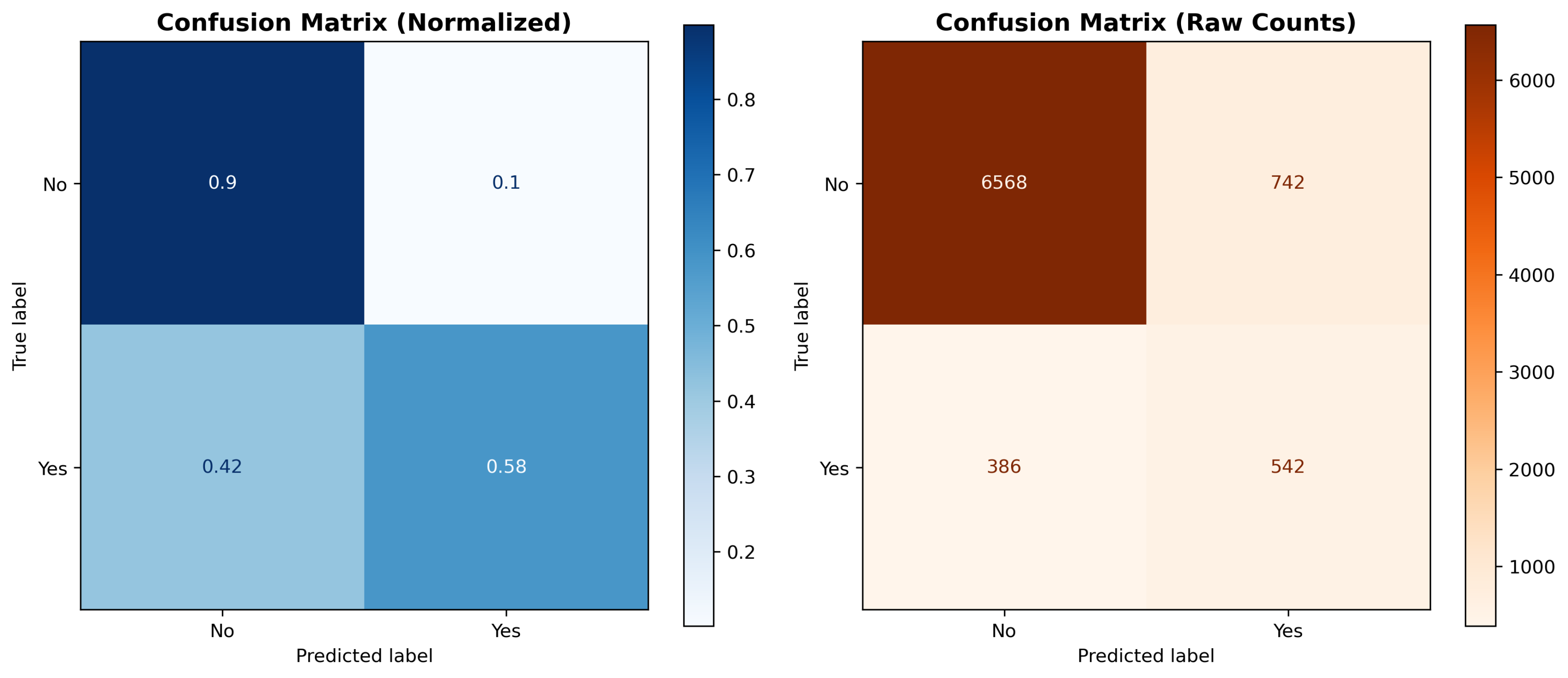
Improvement over Baseline



- F1: +0.207 (+41.3%, $p < 1e-15$)
- AUC: +0.298 (+59.5%, $p < 1e-16$)
- Z-scores: 34.74 (F1), 40.70 (AUC)
→ highly significant

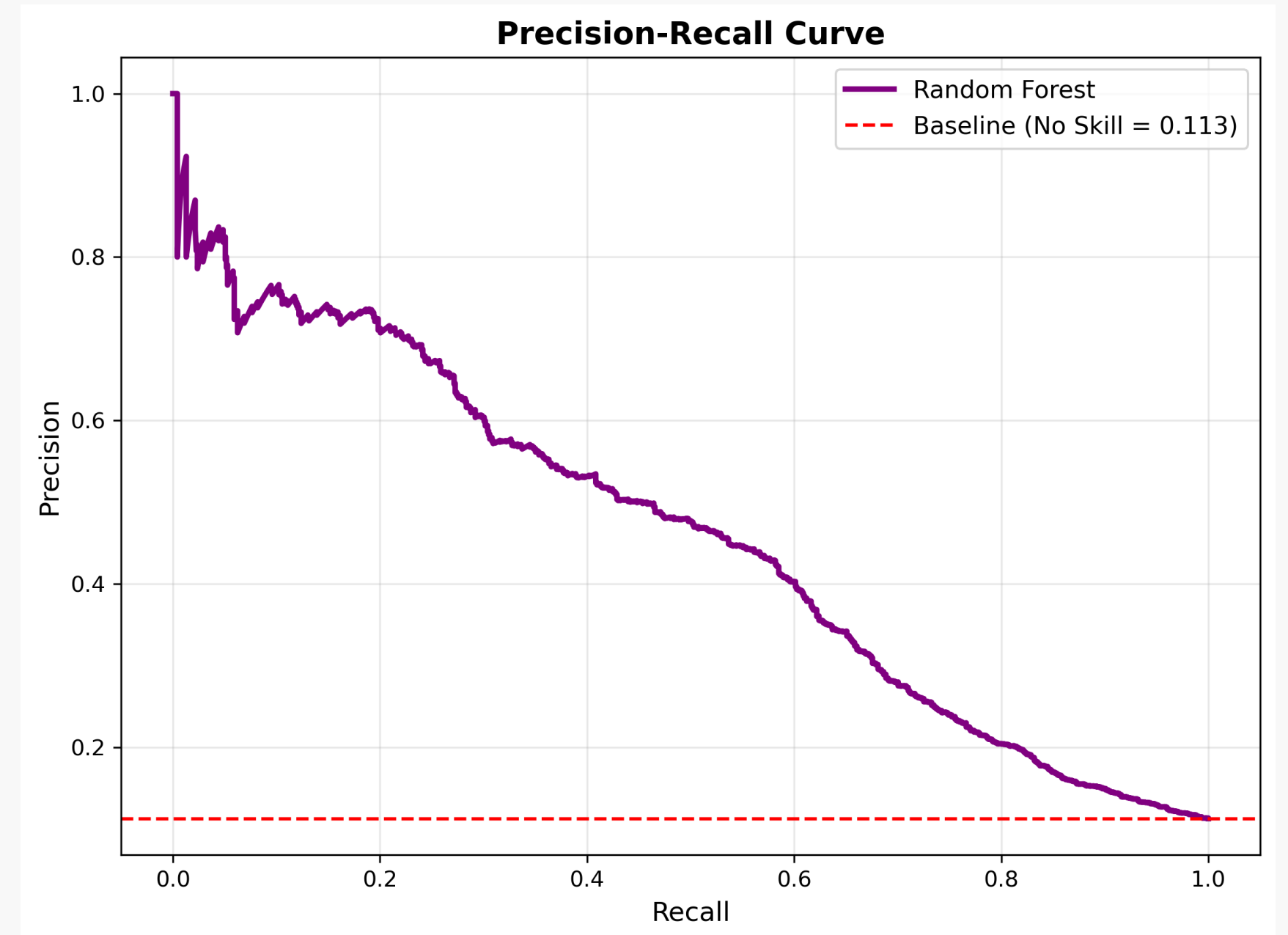
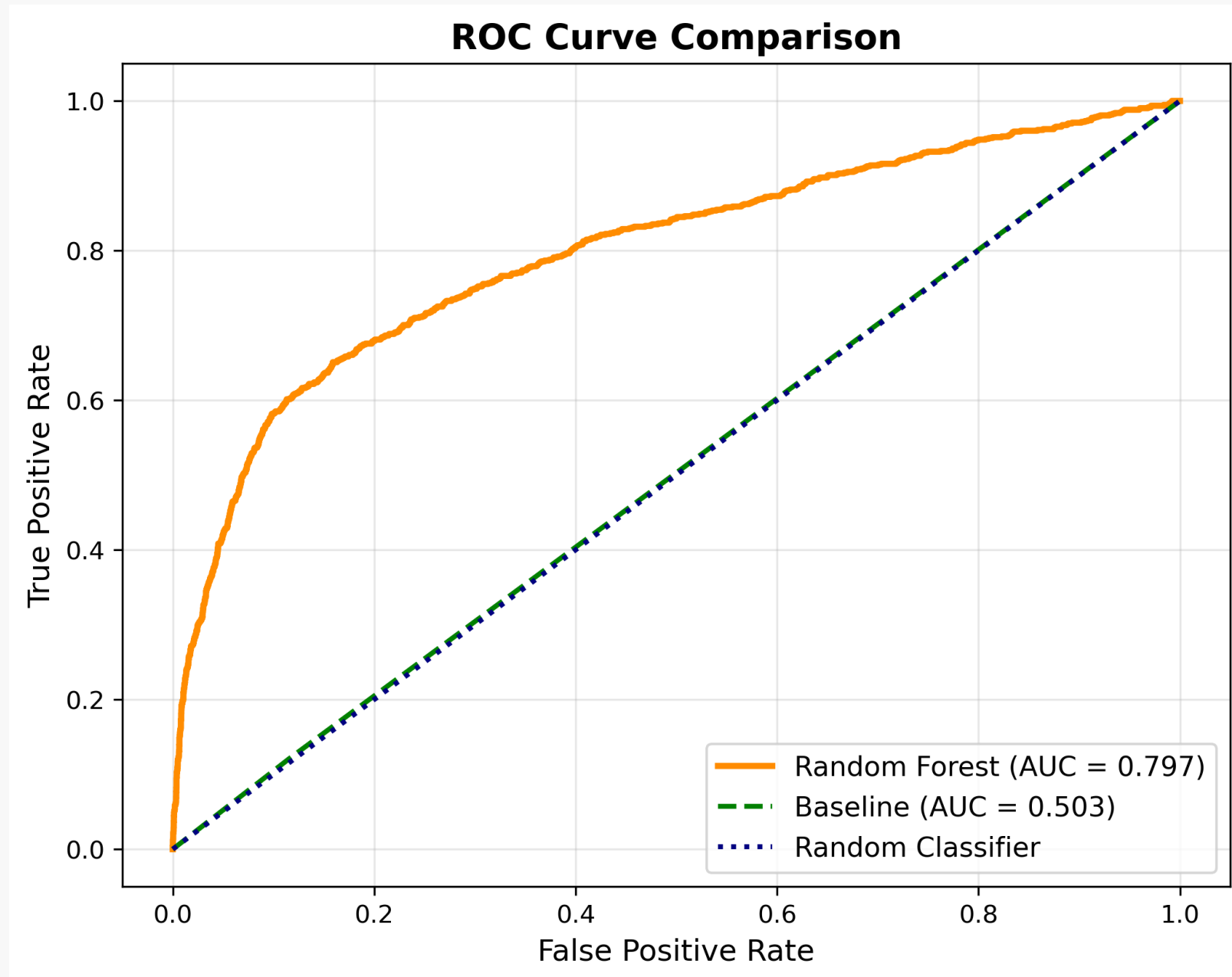
Model Inspection

Confusion Matrix

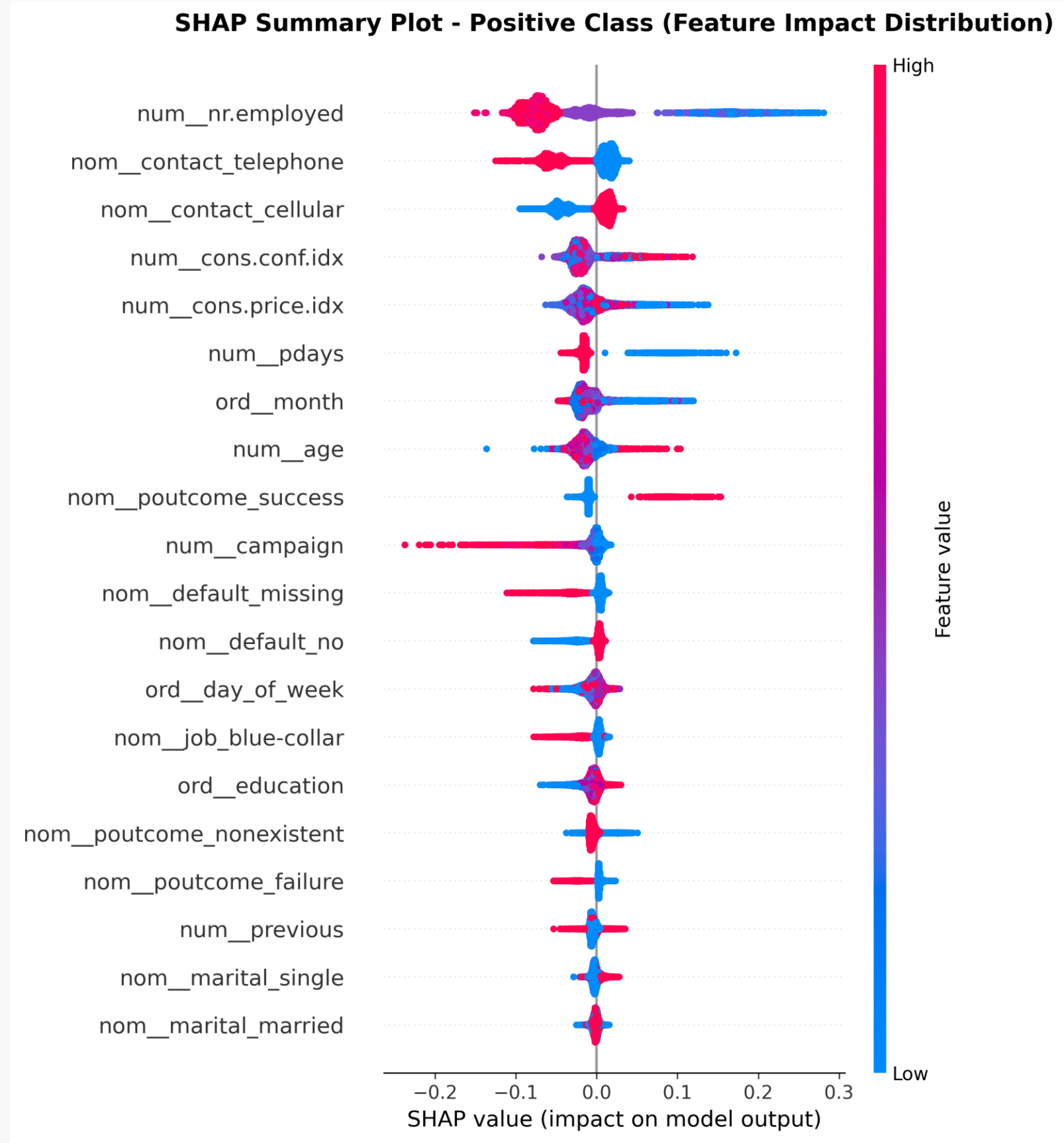


Model Inspection

ROC curve & Precision-Recall Curve



Global Feature Importance

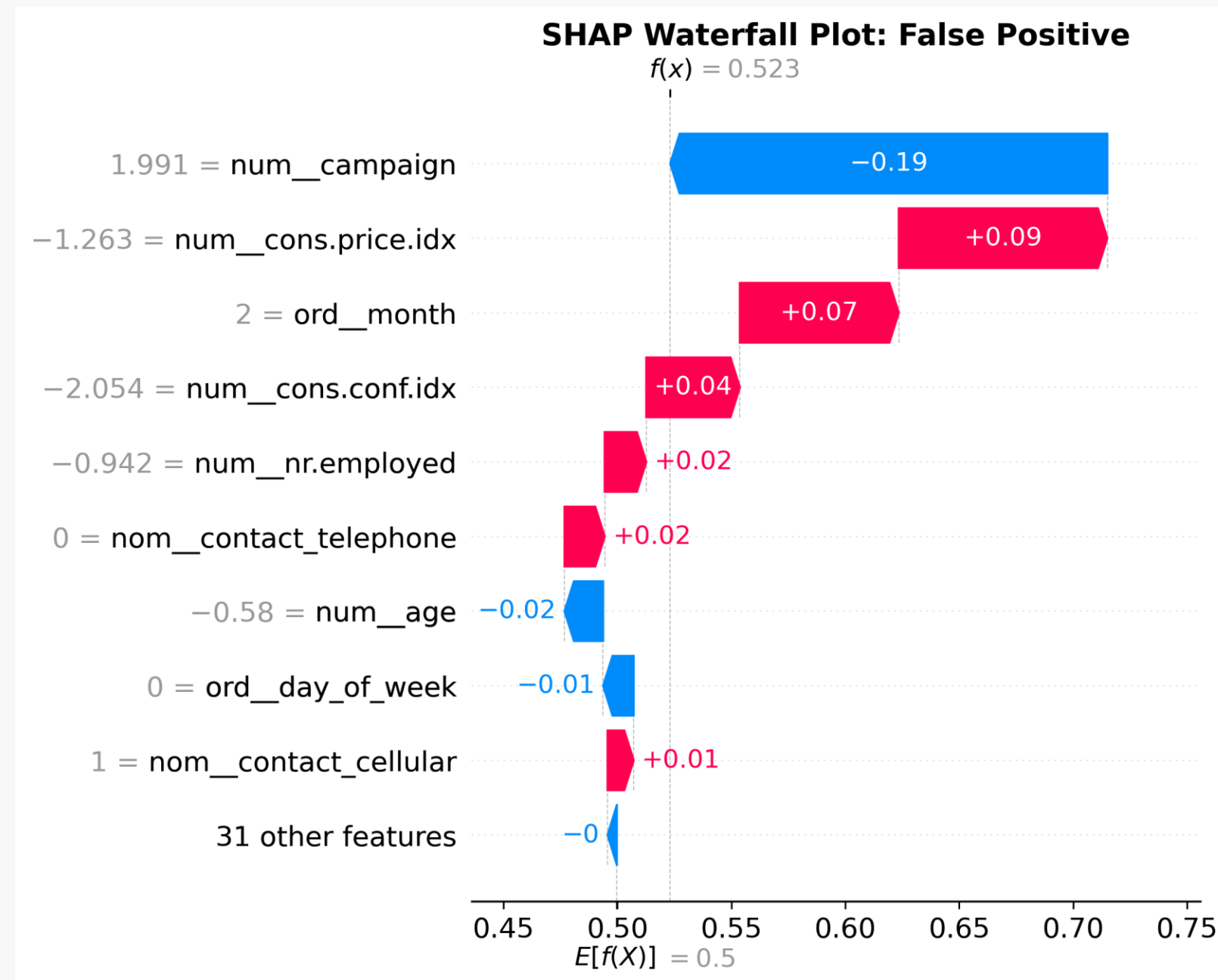


Key insights:

- Top Predictor: Number of employees
 - Low employment → High subscription
- Best Channel: Cellular
- Timing: Recency Matters

Local Feature Importance

False Positive (Incorrectly Predicted "Yes")



- The Conflict: Macro-economics vs. Individual Experience.
- Why Yes? Favorable economic indicators (price.idx, month)
- Why Real Outcome was No? The client was contacted too many times (campaign)

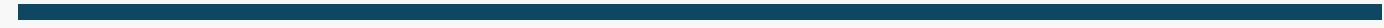
Outlook

Improving Performance

- Feature Eng: Interaction terms (eg. pairs)
- Optimization: Tune Decision Thresholds
- More data: Customer Demographics

Business & Interpretability

- Integration: Marketing Dashboard



Thank you

